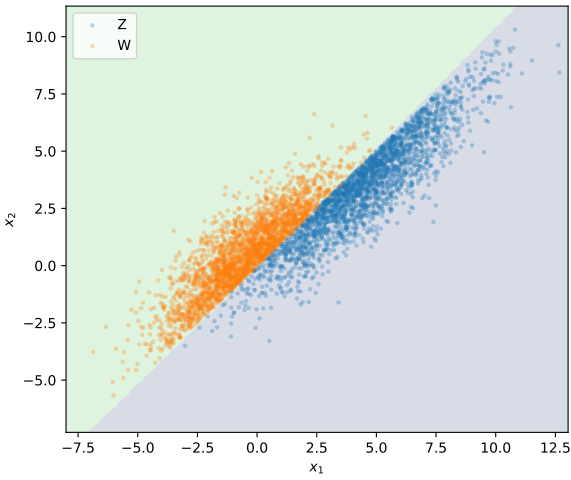


Physics 442: Project 1

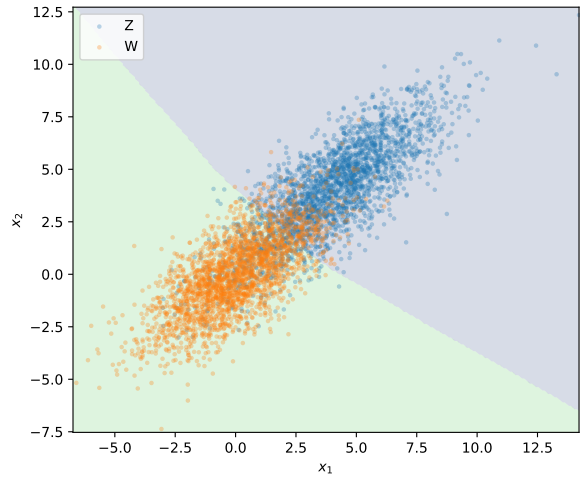
Mingwa Park

September 30, 2026

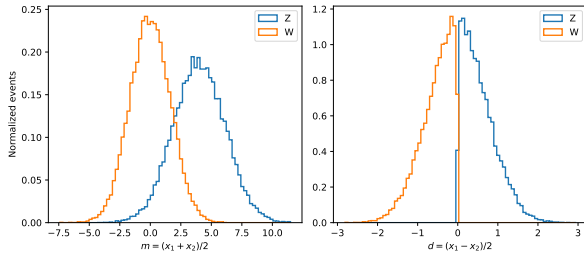
1 WZ Classification



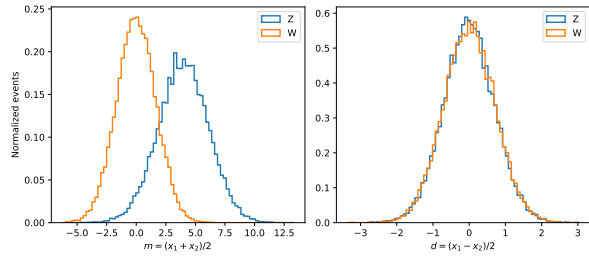
(a) Set 1: classifier decision boundary.



(b) Set 2: classifier decision boundary.



(c) Set 1: distributions of sum and difference of x_1 and x_2



(d) Set 2: distributions of sum and difference of x_1 and x_2

Figure 1: Comparison of the two data sets

A model trained on the dataset 1 implicitly learned the difference of x_1 and x_2 , which is the artifact of post-processing (detector effect). While a model trained on the dataset 2 has worse classification accuracy, but actually this would be better model, especially if we will gonna apply the model to other dataset.

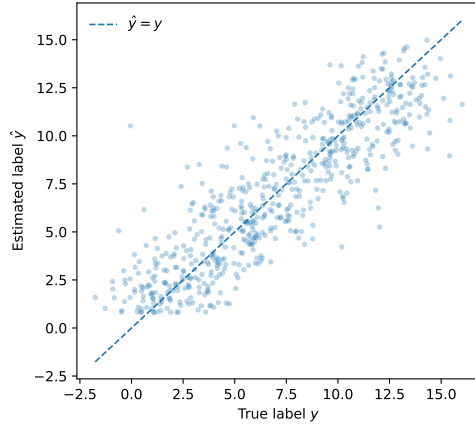
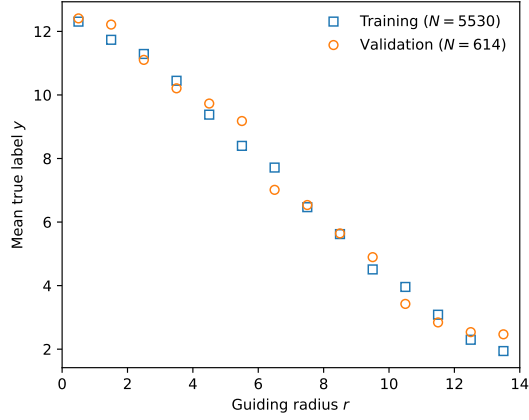


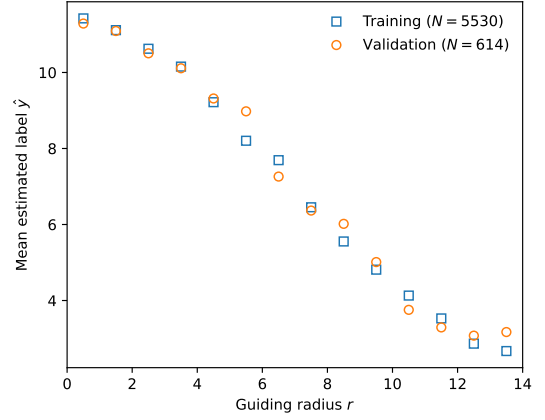
Figure 2: Comparison of true y and estimated $\hat{y}$, on the validation data set.

2 Population-level biases in regression outputs

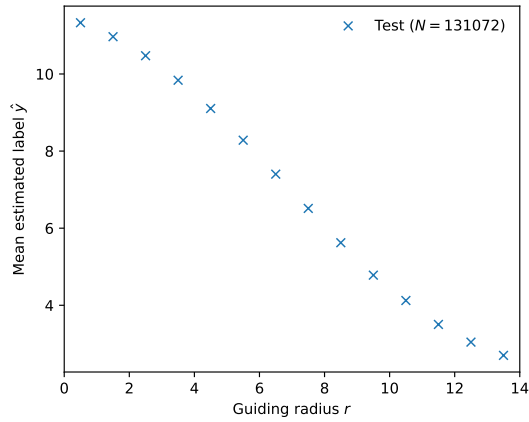
The result of classifier on the validation data set is given in Fig. 2. Other histograms are given in Fig. 3. While the classifier appears to have learned something reasonable, the population level estimates are systematically biased toward the mean, underestimating in the low radius, and overestimating in the high radius. Since the test sample is much larger, it gives much smaller statistical error bars. but the deviation of test set from the true label is larger.



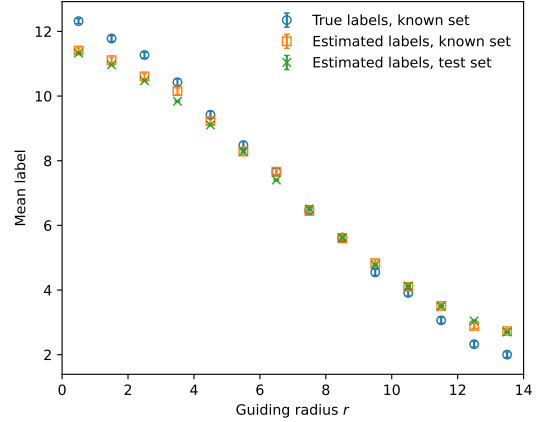
(a) True y to the guiding radius for the training and validation sets.



(b) Estimated $\hat{y}$ to the guiding radius for the training and validation sets.



(c) Estimated $\hat{y}$ on the test set.



(d) Comparison of true and estimated labels for three data sets, with the standard error of each histogram bin.

Figure 3: Mean true and estimated labels as functions of the guiding radius.